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United States Patent	12415362
Kind Code	B2
Date of Patent	September 16, 2025
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Flow path unit and liquid ejection apparatus

Abstract

A flow path unit includes a first flange including a part of a flow path, and a second flange detachably coupled to the first flange and including a part of the flow path, wherein the first flange includes a facing surface that faces the second flange when the second flange is coupled to the first flange, the facing surface is provided with an O-ring annularly surrounding the flow path, a contact surface being in contact with the O-ring annularly surrounding the flow path, and a liquid gasket disposed on the contact surface, and the contact surface is provided with at least one recessed portion.

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Appl. No.:	18/182727
Filed:	March 13, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230294411 A1	Sep. 21, 2023

Foreign Application Priority Data

JP	2022-039997	Mar. 15, 2022
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Publication Classification

Int. Cl.: B41J2/175 (20060101)

U.S. Cl.:

CPC **B41J2/175** (20130101);

Field of Classification Search

CPC: B41J (2/175); B41J (2/17509); B41J (2/1752); B41J (2/17523); B41J (2/18)

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Background/Summary

(1) The present application is based on, and claims priority from JP Application Serial Number 2022-039997, filed Mar. 15, 2022, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to a flow path unit and a liquid ejection apparatus.

2. Related Art

(3) In the related art, for example, in a liquid ejection apparatus or the like, a flow path unit has been used. The flow path unit is configured such that a plurality of members each having a flow path of liquid are coupled to each other by way of an O ring, and the O-ring and a contact surface on which the O-ring is disposed are provided at a coupling portion of the plurality of members. For example, JP-A-5-26381 discloses a seal structure which is configured such that an oil pump housing having an oil passage and a cylinder block having an oil passage are coupled to each other, and an O-ring and a fitting groove into which the O-ring is fitted are provided at a coupling portion between the oil pump housing and the cylinder block.

(4) In the known flow path unit where the O-ring and the contact surface are provided at the coupling portion of the plurality of members, there is a case where a foreign substance enters between the O-ring and the contact surface. When a foreign substance enters between the O-ring and the contact surface, there is a possibility that the liquid leaks from the flow path. Here, in the

seal structure disclosed in JP-5-A-26381, a liquid gasket filling groove is formed around the fitting groove. However, in the seal structure disclosed in Japanese Utility Model Publication No. 5 (1993)-26381, the fitting groove and the liquid gasket filling groove are separately formed at positions separated from each other and hence, in the same manner with the known general flow path unit in which an O-ring and a contact surface are provided at a coupling portion of a plurality of members, when a foreign substance enters between the O-ring and the fitting groove which is the contact surface, there is a possibility that liquid leaks from the flow path.

SUMMARY

(5) A flow path unit according to an aspect of the present disclosure is a flow path unit that includes a flow path configured to supply liquid to an ejection portion that is configured to eject the liquid, the flow path unit including a first flange including a part of the flow path, and a second flange detachably coupled to the first flange and including a part of the flow path, wherein the first flange includes a facing surface that faces the second flange when the second flange is coupled to the first flange, the facing surface is provided with an O-ring annularly surrounding the flow path, a contact surface being in contact with the O-ring annularly surrounding the flow path, and a liquid gasket disposed on the contact surface, and the contact surface is provided with at least one recessed portion.

(6) A liquid ejection apparatus according to another aspect of the present disclosure is a liquid ejection apparatus that includes an ejection portion configured to eject liquid, and a flow path configured to supply the liquid to the ejection portion, the liquid ejection apparatus including a first flange including a part of the flow path, and a second flange detachably coupled to the first flange and including a part of the flow path, wherein the first flange includes a facing surface that faces the second flange when the second flange is coupled to the first flange, the facing surface is provided with an O-ring annularly surrounding the flow path, a contact surface being in contact with the O-ring annularly surrounding the flow path, and a liquid gasket disposed on the contact surface, and the contact surface is provided with at least one recessed portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic perspective view illustrating a liquid ejection apparatus according to a first embodiment of the present disclosure.
- (2) FIG. 2 is a perspective view illustrating a flow path unit of the liquid ejection apparatus in FIG. 1.
- (3) FIG. 3 is a perspective view illustrating a carriage-side flange of the flow path unit of the liquid ejection apparatus in FIG. 1.
- (4) FIG. 4 is a perspective view illustrating an ink-cartridge-side flange of the flow path unit of the liquid ejection apparatus in FIG. 1.
- (5) FIG. 5 is a perspective view illustrating an O-ring of the flow path unit of the liquid ejection apparatus in FIG. 1.
- (6) FIG. 6 is a schematic cross-sectional view illustrating a coupling portion between the carriage-side flange and the ink-cartridge-side flange of the flow path unit of the liquid ejection apparatus in FIG. 1.
- (7) FIG. 7 is a schematic cross-sectional view illustrating a state where a foreign substance enters the coupling portion in FIG. 6.
- (8) FIG. 8 is a schematic cross-sectional view illustrating a preferred depth of a concave portion of the flow path unit of the liquid ejection apparatus in FIG. 1.
- (9) FIG. 9 is a schematic cross-sectional view illustrating a preferred pitch of the recessed portions of the flow path unit of the liquid ejection apparatus in FIG. 1.

(10) FIG. 10 is a schematic cross-sectional view illustrating a coupling portion between a carriage-side flange and an ink-cartridge-side flange of a flow path unit of a liquid ejection apparatus according to a second embodiment of the present disclosure.

(11) FIG. 11 is a schematic cross-sectional view illustrating a coupling portion between a carriage-side flange and an ink-cartridge-side flange of a flow path unit of a liquid ejection apparatus of a reference example.

(12) FIG. 12 is a schematic cross-sectional view illustrating a state where a foreign substance enters the coupling portion in FIG. 11.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

(13) Hereinafter, the present disclosure will be schematically described.

(14) A flow path unit according to a first aspect of the present disclosure is a flow path unit that includes a flow path configured to supply liquid to an ejection portion that is configured to eject the liquid, the flow path unit including a first flange including a part of the flow path, and a second flange detachably coupled to the first flange and including a part of the flow path, wherein the first flange includes a facing surface that faces the second flange when the second flange is coupled to the first flange, the facing surface is provided with an O-ring annularly surrounding the flow path, a contact surface being in contact with the O-ring annularly surrounding the flow path, and a liquid gasket disposed on the contact surface, and the contact surface is provided with at least one recessed portion.

(15) According to this aspect, in the facing surface, the liquid gasket is disposed on the contact surface on which the O-ring is disposed, and at least one recessed portion is provided on the contact surface. Accordingly, when a foreign substance enters between the O-ring and the contact surface, the liquid gasket can wrap the foreign substance via the recessed portion. That is, it is possible to suppress the occurrence of a gap in the facing surface due to entry of a foreign substance between the O-ring and the contact surface. Accordingly, even in a case where a foreign substance enters between the O-ring and the contact surface, it is possible to bring about a state where the liquid gasket wraps around the foreign substance and hence, the lowering of sealing performance at the facing surface can be suppressed.

(16) A flow path unit according to a second aspect of the present disclosure is characterized in that, in the flow path unit according to the first aspect, the O-ring is provided with an annular groove annularly surrounding the flow path.

(17) According to this aspect, the O-ring is provided with the annular groove that annularly surrounds the flow path. Accordingly, the O-ring can accommodate the liquid gasket in the annular groove, and a large amount of liquid gasket can be disposed in the annular groove. Therefore, it is possible to particularly effectively suppress the lowering of the sealing performance at the facing surface.

(18) A flow path unit according to a third aspect of the present disclosure is characterized in that, in the flow path unit according to the first or second aspect, at least one of the recessed portions annularly surrounds the flow path.

(19) According to this aspect, at least one of the recessed portions annularly surrounds the flow path. With such a configuration, even when a foreign substance enters from any direction on the facing surface, it is possible to bring about a state where the liquid gasket wraps around the foreign substance and hence, the lowering of the sealing performance on the facing surface can be suppressed.

(20) A liquid ejection apparatus according to a fourth aspect of the present disclosure is a liquid ejection apparatus that includes an ejection portion configured to eject liquid, and a flow path configured to supply the liquid to the ejection portion, the liquid ejection apparatus including a first flange including a part of the flow path, and a second flange detachably coupled to the first flange and including a part of the flow path, wherein the first flange has a facing surface that faces the second flange when the second flange is coupled to the first flange, the facing surface is provided

with an O-ring annularly surrounding the flow path, a contact surface being in contact with the O-ring annularly surrounding the flow path, and a liquid gasket disposed on the contact surface, and the contact surface is provided with at least one recessed portion.

(21) According to this aspect, in the facing surface, the liquid gasket is disposed on the contact surface on which the O-ring is disposed, and at least one recessed portion is provided on the contact surface. Accordingly, even in a case where a foreign substance enters between the O-ring and the contact surface, it is possible to bring about a state where the liquid gasket wraps around the foreign substance and hence, it is possible to eject the liquid while suppressing the lowering of sealing performance at the facing surface.

First Example

(22) Hereinafter, a liquid ejection apparatus provided with a flow path unit according to one embodiment of the present disclosure will be described in detail with reference to attached drawings. First, the liquid ejection apparatus according to a first embodiment of the present disclosure is schematically described.

(23) The liquid ejection apparatus **1** according to the present embodiment is configured to perform recording on a medium **P** transported in a transport direction **A** by a transport unit not illustrated in the drawing by moving a carriage **2** provided with an ejection unit **3** configured to eject ink on a side opposite to the medium **P** in a reciprocating manner in a scanning direction **B** intersecting with the transport direction **A**. To be more specific, the medium **P** is intermittently driven in the transport direction **A**, the ejection unit **3** is moved in a reciprocating manner (scanned in a reciprocating manner) in the scanning direction **B** via the carriage **2**, and ink is ejected from a plurality of nozzles not illustrated in the drawing formed in the ejection unit **3** thus performing recording. Here, one direction of the scanning directions **B** is referred to as a direction **B1**, and the other direction of the scanning directions **B** is referred to as a direction **B2**.

(24) The ejection unit **3** of the present embodiment is an inkjet head including a piezoelectric element. However, the present disclosure is not limited to such a configuration, and a thermal head including a heater may be used instead of the piezoelectric element. Further, the carriage **2** is coupled to ink cartridges **4** that respectively store black ink, cyan ink, magenta ink, and yellow ink by way of ink tubes **5** that constitute a part of the flow path unit **100** and also constitute flow paths **50** for inks. The flow path unit **100** is provided on a direction **B1** side of the carriage **2**, and the detailed configuration of the flow path unit **100** will be described later. Further, the ejection unit **3** is configured to eject black ink, cyan ink, magenta ink, and yellow ink, and nozzle rows corresponding to the respective inks are formed along the transport direction **A**. The number of colors of ink is not limited to these four colors, and may be more or less than four colors, or other colors may be adopted.

(25) Next, a detailed configuration of the flow path unit **100** (flow path unit **100A**) in the liquid ejection apparatus **1** according to the present embodiment will be described with reference to FIG. **2** to FIG. **9**, FIG. **11**, and FIG. **12**. As illustrated in FIG. **2**, the flow path unit **100A** according to the present embodiment includes a tube-side flange member **120** coupled to the ink tubes **5**, and a carriage-side flange member **110**. Then, the tube-side flange member **120** and the carriage-side flange member **110** are coupled to each other, and the carriage-side flange member **110** is fixed to the carriage **2** by way of a fixing member **140**.

(26) Here, FIG. **3** illustrates a side of the carriage-side flange member **110** facing the tube-side flange member **120**, and FIG. **4** illustrates a side of the tube-side flange member **120** facing the carriage-side flange member **110**. As illustrated in FIG. **3**, the carriage-side flange member **110** has four cylindrical portions **111** constituting a part of a flow path **50**, a planar portion **112** positioned around the cylindrical portions **111**, and a convex portion **114** positioned around the planar portion **112**. The planar portion **112** is provided with a plurality of recessed portions **113** annularly surrounding the cylindrical portion **111**. On the other hand, as illustrated in FIG. **4**, the tube-side flange member **120** has four hole portions **121** constituting a part of the flow path **50**, and a

pedestal-shaped planar portion **122** positioned around the hole portions **121**. The planar portion **122** is provided with a plurality of recessed portions **123** annularly surrounding the hole portion **121**. The four cylindrical portions **111** and the four hole portions **121** respectively correspond to four inks of black ink, cyan ink, magenta ink, and yellow ink.

(27) In the present embodiment, the recessed portion **113** and the recessed portion **123** have substantially the same configuration. However, a recessed portion provided to the carriage-side flange member **110** and a recessed portion provided to the tube-side flange member **120** may have different configurations, or a configuration may be adopted where a recessed portion is provided to only one of the carriage-side flange member **110** and the tube-side flange member **120**.

(28) The carriage-side flange member **110** and the tube-side flange member **120** are coupled to each other at a position where the cylindrical portions **111** are inserted into the hole portions **121** and the planar portion **112** and the planar portion **122** face each other. When the carriage-side flange member **110** and the tube-side flange member **120** are coupled to each other, a quadruple O-ring **130** illustrated in FIG. 5 is disposed at a position between the planar portion **112** and the planar portion **122**. The O-ring **130** has convex portions **131**, since the convex portion **131** is pressed by the planar portion **112** and the planar portion **122**, the carriage-side flange member **110** and the tube-side flange member **120** are sealed. Here, although the O-ring **130** according to the present embodiment has a quadruple ring configuration, the O-ring **130** is not limited to such a configuration.

(29) Further, in the flow path unit **100A** of the present embodiment, a liquid gasket **150** is disposed at a position where the planar portion **112** and the planar portion **122** face each other, that is, a position around the O-ring **130**. Here, FIG. 6 and FIG. 7 are schematic cross-sectional views illustrating a position between the planar portion **112** and the planar portion **122** in an enlarged manner. As illustrated in FIG. 6 and FIG. 7, recessed portions **113** and recessed portions **123** are provided at positions of the planar portion **112** and the planar portion **122** facing the O-ring **130**, and the liquid gasket **150** is disposed also in the recessed portions **113** and the recessed portions **123**. By disposing the liquid gasket **150** in this manner, for example, even when a foreign substance O enters between the O-ring **130** and the planar portion **112** or the like that is a contact surface with the O-ring **130** as illustrated in FIG. 7, it is possible to bring about a state where the liquid gasket **150** wraps the foreign substance O and hence, the lowering of the sealing performance at the facing surface **115** of the carriage-side flange member **110** and at the facing surface **125** of the tube-side flange member **120** can be suppressed. In the flow path unit **100A** of the present embodiment, the facing surface **115** is constituted of the planar portion **112** and the convex portion **114**, and the facing surface **125** is constituted of the planar portion **122**.

(30) Here, FIG. 11 and FIG. 12 are schematic cross-sectional views illustrating a position between a planar portion **112** and a planar portion **122** of a flow path unit **101** of a liquid ejection apparatus of a reference example in an enlarged manner. The liquid ejection apparatus of the reference example has substantially the same configuration as the flow path unit **100A** of the present embodiment except that the planar portion **112** and the planar portion **122** have no recessed portion, and the liquid gasket is not disposed on a facing surface **115** of a carriage-side flange member **110** and a facing surface **125** of a tube-side flange member **120**. With such a configuration, in FIG. 11 and FIG. 12 constitutional members corresponding to those of the flow path unit **100A** of the present embodiment are expressed by giving the same symbols.

(31) As illustrated in FIG. 11 and FIG. 12, the flow path unit **101** of the liquid ejection apparatus of the reference example has no recessed portion in the planar portion **112** and the planar portion **122**, and the liquid gasket is not disposed on the facing surface **115** of the carriage-side flange member **110** and the facing surface **125** of the tube-side flange member **120**. Accordingly, as illustrated in FIG. 12, when a foreign substance O enters between the O-ring **130** and the planar portion **112** or the like that is a contact surface with the O-ring **130**, a gap G is generated. Therefore, the sealing performance at the facing surface **115** of the carriage-side flange member **110** and the facing

surface **125** of the tube-side flange member **120** is lowered. When the sealing performance at the facing surface **115** of the carriage-side flange member **110** and the facing surface **125** of the tube-side flange member **120** is lowered, there is a possibility that the ink leaks out from the flow path **50** through the gap G.

(32) As described above, the flow path unit **100A** according to the present embodiment is the flow path unit **100** including the flow path **50** for supplying ink to the ejection unit **3** configured to eject ink that is liquid. Further, the flow path unit **100A** includes a carriage side flange member **110** as a first flange having a part of the flow path **50** and a tube-side flange member **120** as a second flange detachably coupled to the carriage-side flange member **110** and having a part of the flow path **50**. Here, the carriage-side flange member **110** has a facing surface **115** that faces the tube-side flange member **120** when the tube-side flange member **120** is coupled to the carriage-side flange member **110**, on the facing surface **115**, an O-ring **130** that annularly surrounds the flow path **50**, a planar portion **112** as a contact surface with which the O-ring **130** is brought into contact in a state of annularly surrounding the flow path **50**, and a liquid gasket **150** disposed on the planar portion **112** are provided, and at least one recessed portion **113** is provided to the planar portion **112**.

(33) In the flow path unit **100A** of the present embodiment, in the above-mentioned description, the tube-side flange member **120** may be regarded as a first flange, and the carriage-side flange member **110** may be regarded as a second flange. To be more specific, the flow path unit **100A** of the present embodiment includes the tube-side flange member **120** as a first flange having a part of the flow path **50**, and the carriage-side flange member **110** as a second flange detachably coupled to the tube-side flange member **120** and having a part of the flow path **50**. Here, the tube-side flange member **120** has a facing surface **125** that faces the carriage-side flange member **110** when the carriage-side flange member **110** is coupled to the tube-side flange member **120**, on the facing surface **125**, an O-ring **130** that annularly surrounds the flow path **50**, a planar portion **122** as a contact surface with which the O-ring **130** is brought into contact in a state of annularly surrounding the flow path **50**, and a liquid gasket **150** disposed on the planar portion **112** are provided, and at least one recessed portion **123** is provided to the planar portion **122**.

(34) As described above, in the facing surface (at least one of the facing surfaces **115** and **125**), the liquid gasket **150** is disposed on the contact surface (at least one of the planar portion **112** and the planar portion **122**) on which the O-ring **130** is disposed, and at least one recess is provided to the contact surface. Accordingly, when a foreign substance O enters between the O-ring **130** and the contact surface, the liquid gasket **150** can wrap the foreign substance O via the recessed portion. That is, it is possible to suppress the occurrence of a phenomenon that the gap G is generated at the facing surface due to entry of the foreign substance O between the O-ring **130** and the contact surface. Accordingly, even in a case where a foreign substance O enters between the O-ring **130** and the contact surface, it is possible to bring about a state where the liquid gasket **150** wraps around the foreign substance O and hence, the lowering of the sealing performance at the facing surface can be suppressed. Further, for example, even in a case where the O-ring **130** is deformed due to change with time or the like, it is possible to bring about a state where the liquid gasket **150** wraps around a deformed portion of the O-ring **130** and hence, it is possible to suppress the lowering of the sealing performance at the facing surface.

(35) Here, describing from the viewpoint of the liquid ejection apparatus, the liquid ejection apparatus **1** according to the present embodiment is a liquid ejection apparatus including the flow path unit **100** having the above-described characteristics, the ejection unit **3** configured to eject ink that is liquid, and the flow path **50** configured to supply ink to the ejection unit **3**. Accordingly, even in a case where a foreign substance O enters between the O-ring **130** and the contact surface, it is possible to bring about a state where the liquid gasket **150** wraps around the foreign substance O and hence, it is possible to eject the liquid while suppressing the lowering of the sealing performance at the facing surface.

(36) Here, as illustrated in FIG. 3, in the flow path unit **100A** of the present embodiment, the

recessed portions **113** annularly surround the cylindrical portion **111** constituting the flow path **50**. Further, as illustrated in FIG. **4**, in the flow path unit **100A** of the present embodiment, the recessed portions **123** annularly surround the hole portion **121** constituting the flow path **50**. As described above, it is preferable that at least one of the recessed portion **113** and the recessed portion **123** annularly surround the flow path **50**. With such a configuration, even when a foreign substance enters from any direction on the facing surface, it is possible to bring about a state where the liquid gasket **15** wraps around the foreign substance **O** and hence, the lowering of the sealing performance on the facing surface can be suppressed. However, the present disclosure is not limited to such a configuration. For example, a configuration may not be adopted where at least one of the plurality of recessed portions **113** and the plurality of recessed portions **123** annularly surround the flow path **50**, and further, a configuration may be adopted where none of the plurality of recessed portions **113** and the plurality of recessed portions **123** annularly surround the flow path **50**.

(37) With respect to a preferable configuration of the recessed portion **113** and the recessed portion **123**, for example, as illustrated in FIG. **8**, assuming a depth of the recessed portion **113** and the recessed portion **123** as $L2$, and assuming a sinking amount of the O-ring **130** into the recessed portion **113** and the recessed portion **123** as $L1$, it is preferable that $L2$ be set to be larger than $L1$, that is, the relationship of $L2 > L1$ be established. This is because a region for accommodating the liquid gasket **150** can be suitably ensured between the O-ring **130** and the recessed portion **113** and the recessed portion **123**.

(38) Further, as illustrated in FIG. **9**, for example, when a plurality of recessed portions **113** and the plurality of recessed portions **123** are formed in the contact surfaces, assuming a forming pitch of the recessed portions **113** and the recessed portions **123** as $L4$, and a contact length of the O-ring **130** with respect to the contact surface as $L3$, it is preferable that $L3$ and $L4$ be set such that $L4$ is smaller than a half of the length $L3$, that is, a relationship of $L4 < (L3/2)$ be satisfied. This is because a region for accommodating the liquid gasket **150** can be suitably ensured between the O-ring **130** and the recessed portion **113** and the recessed portion **123**.

Second Embodiment

(39) Hereinafter, a flow path unit **100B** of a liquid ejection apparatus according to a second embodiment will be described with reference to FIG. **10**. Here, FIG. **10** is a view corresponding to FIG. **6** in the liquid ejection apparatus **1** according to the first embodiment. Further, the liquid ejection apparatus of the present embodiment is substantially the same as the liquid ejection apparatus of the first embodiment except for a configuration described hereinafter and hence, the liquid ejection apparatus of the present embodiment has substantially the same characteristics as the liquid ejection apparatus of the first embodiment. Here, in FIG. **10**, constitutional elements common to the above-described first embodiment are given the same symbols, and their detailed descriptions will be omitted.

(40) As illustrated in FIG. **5** and FIG. **6**, in the flow path unit **100A** of the first embodiment, the O-ring **130A** had the convex portions **131**, and there is no recess or the like in the convex portions **131**. On the other hand, as illustrated in FIG. **10**, in the flow path unit **100B** of the present embodiment, a recess **132** is formed in the convex portion **131**. Specifically, in the flow path unit **100B** of the present embodiment, an O-ring **130B** has the recess **132** formed in the convex portion **131**, and the recess **132** is an annular groove that annularly surrounds a flow path **50**. Accordingly, in the flow path unit **100B** of the present embodiment, the O-ring **132** can accommodate a liquid gasket **150** in the recess **132** that is an annular groove, and a large amount of liquid gasket **150** can be disposed in the recess **132**. Therefore, it is possible to particularly effectively suppress the lowering of the sealing performance at the facing surfaces **115** and **125**. Further, a large amount of liquid gasket **150** can be disposed in the recess **132** that is an annular groove and hence, it is also possible to suppress the flow-out of the liquid gasket **150** to an unintended place.

(41) Note that the disclosure is not limited to the aforementioned example, and many variations are

possible within the scope of the disclosure as described in the appended claims. It goes without saying that such variations also fall within the scope of the disclosure.

Claims

1. A flow path unit comprising: a flow path configured to supply liquid to an ejection unit that is configured to eject the liquid, a first flange including a part of the flow path, and a second flange detachably coupled to the first flange and including a part of the flow path, wherein the first flange includes a facing surface that faces the second flange when the second flange is coupled to the first flange, the facing surface is provided with an O-ring annularly surrounding the flow path, a contact surface being in contact with the O-ring annularly surrounding the flow path, and a liquid gasket disposed on the contact surface, and the contact surface is provided with at least one recessed portion that is part of the contact surface that contacts the O-ring.
 2. The flow path unit according to claim 1, wherein the O-ring is provided with an annular groove annularly surrounding the flow path.
 3. The flow path unit according to claim 1, wherein at least one of a plurality of the recessed portions annularly surrounds the flow path.
 4. A liquid ejection apparatus comprising: an ejection unit configured to eject liquid; and a flow path configured to supply the liquid to the ejection unit, the liquid ejection apparatus including a first flange including a part of the flow path, and a second flange detachably coupled to the first flange and including a part of the flow path, wherein the first flange includes a facing surface that faces the second flange when the second flange is coupled to the first flange, the facing surface is provided with an O-ring annularly surrounding the flow path, a contact surface being in contact with the O-ring annularly surrounding the flow path, and a liquid gasket disposed on the contact surface, and the contact surface is provided with at least one recessed portion that is part of the contact surface that contacts the O-ring.
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